High Integrity GNSS Navigation and Safe Separation Distance to Support Local-Area UAV Networks

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ABSTRACT

This paper develops an methodology to estimate safe separation distance for UAVs which share the same source of guidance and LADGNSS differential corrections.

1.0. INTRODUCTION

Recent technological improvements and increased military utilization have proven the operational viability of unmanned aerial vehicles (UAVs) and made them attractive for a wide range of potential civil applications. In particular, the capability and low cost of small UAVs have led to a dramatic growth in potential commercial applications over the past few years as well as a major interest in the media. While most current uses of UAVs are remotely piloted by humans, future UAVs will operate autonomously and coordinate their activities in large groups. Autonomous UAVs networks guided by central controllers will safely and routinely perform various missions such as surveillance, reconnaissance, and data collection [1].

Prior work [1, 2] has proposed a concept of UAV network operations that use differential corrections from Local-Area Differential GNSS (LADGNSS) to provide increased accuracy, safety, and reliable navigation to UAVs within 5 to 100 km of the controller station. LADGNSS supports low cost for commercial applications and high integrity to allow UAVs to operate in close proximity to each other while minimizing collision risk. Local-area UAV networks can perform their tasks safely as long as UAV locations within the network are monitored by the controller to insure that safe separation distances are maintained from other vehicles and ground obstructions. The controller, including a LADGNSS reference station, receives GNSS signals and positions of UAVs, estimates errors and separation distances, and broadcasts differential corrections and optimized guidance for precise and safe operation of UAVs.

This paper presents a methodology to estimate the desired minimum separation distance that should be defined to prevent collisions by being utilized for guidance of UAVs. Separation distance can be derived by modeling mainly two sources of position uncertainty that could potentially lead to collisions: (i) navigation error, or the residual error in the estimated position (compared with the true position), which cannot be completely eliminated by LADGNSS; and (ii) flight technical error, or error in following the path defined by the guidance source, which depends on the controller's performance and external disturbances such as wind.

This paper estimates these errors using standard error models of LADGNSS developed for manned aircraft and bounds derived from empirical data collected from UAV flight tests.

2.0. LOCAL AREA UAV NETWORK OVERVIEW

Figure 1 illustrates the concept of local-area UAV network operations developed in [1,2]. The control station shown at the lower left is the source of local-area differential GNSS corrections and integrity information as well as real-time guidance for each UAV in the network. The LADGNSS and guidance information are separate data messages combined in the same outbound transmission. The guidance function also requires feedback from each UAV in real time, including its current position and velocity. Therefore, a two-way datalink is required and is used to relay GNSS information (such as position-domain protection levels) as well as position and velocity from UAVs to the control station.

Safe separation distance

3.0. MODLELING(?) METHODOLOGY

Theoretically experimentally

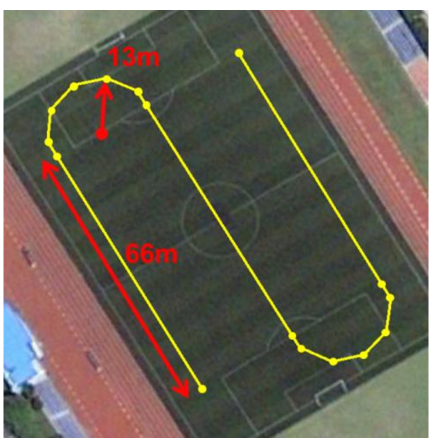
3.1. RESIDUAL TROPOSPHERIC ERROR MODLEING

3.2. RESIDUAL IONOSPHERIC ERROR MODLEING

3.3. AIRBORNE MULTIPATH MODLEING

3.4. FLIGHT TECHNICAL ERROR ESTIMATION

In order to estimate flight technical error (FTE), firstly, it is required to set-up path-following guidance for UAV. In this research, flight path is designed to cover interesting area considering practical applications of UAV such as agriculture, surveying, mapping, and reconnaissance. Figure (?) shows designed flight path. The path consists of three straight sections and two semicircular sections. The ground altitude of each waypoint is fixed as 10m.



UAV flight test is conducted along the path, and flight trajectory information is logged at 10 hertz during the test. ??

Lateral and vertical FTE are calculated by subtracting designed guidance from logged position in each direction. However, the entire FTE data cannot be used to estimate statistical characteristics, because assumption that each error samples are independent is invalid. Consequently, use of the entire error samples can causes significant autocorrelation in the FTE data. Therefore, resampling FTE points along flight path with constant distance interval is required [10]. First, in order to evaluate the significance of autocorrelation at kth lag for N samples, the autocorrelation function is defined as follows:

$$\gamma_k = \frac{\sum_{i=1}^{N-k}(x_i - \bar{x})(x_{i+k} - \bar{x})}{\sum_{i=1}^{N}(x_i - \bar{x})^2}$$

where x_i is ith data sample and $\bar{x}$ is mean of data samples. The significance of autocorrelation at the kth lag is tested to determine the distance interval between samples. The null hypothesis that kth lag auto correlation value is statistically zero can be expressed as:

$$H_0 : \rho_k = 0$$

The hypothesis is evaluated with the critical values for type I error level α .

$$\gamma_{\alpha,k} = N^{-1}(\alpha,0,1)\sqrt{\frac{1}{n}\left(1+2\sum_{i=1}^{k}\gamma_i^2\right)}$$

where $k = 1,\cdots,\mathrm{int}(N/4)$ and $N^{-1}(\alpha,0,1)$ is standard normal variate of standard normal inverse CDF at α . The null hypothesis at kth lag is rejected if $\gamma_k > \gamma_{k,\alpha}$.

4.0. CONFIGURATION

UAV Specification	
Platform	Octo-rotor
Diameter	110 cm
Height	35 cm
Weight	5.31 kg
Flight Time	10 min
Battery	LiPo 6s 22.2V, 5000mAh
Payloads	
GPS Antenna	*NovAtel ProPak-V3*
GPS Receiver	*NovAtel ???*
Flight Controller	*Ardupilot Mega 2.6*
Modem for DGPS	*Xbee Pro Series 1*

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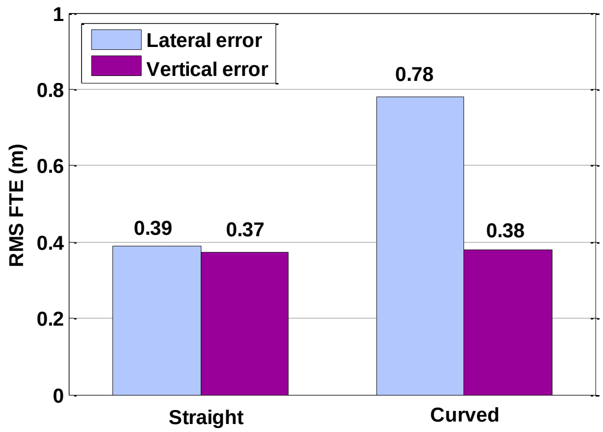
5.0. RESULT

5.1. AIRBORNE MULTIPATH MODLEING

5.2. FLIGHT TECHNICAL ERROR MODLEING

Total 21 of path-following flight are conducted, and 51236 of FTE samples are collected. Following the resampling process noted in preceding section, sampling distance is determined as 5.15m when type I error level α is 99.5%. As a result of the resampling, only 1239 of FTE samples are left.



FTE error data are categorized into FTE in straight segment and FTE in curved segment. Figure (?) shows that lateral FTE RMS error in curved segment is much greater than other FTE errors. This is because the lateral FTE in curved segment is directly influenced by the momentum of the UAV. In addition, the UAV maneuvers on the horizontal plane more frequently when it makes turn around than when it just go straight, so it can also increase lateral FTE. On the other hand, the difference between the vertical FTEs of straight segment and curved segment is very small. This is because there is no variation in vertical guidance.

5.3. SEPARATION DISTANCE

ACKNOWLEDGMENTS

REFERENCES

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[2] D

[3] D

[4] D

[5] d

[6] D

[7] D

[8] D

[9] D

[10] Levy, B.S., Som, P., Greenhaw, R., "Analysis of Flight Technical Error on Straight, Final Approach Segments," *Proceedings of the 59th Annual Meeting of The Institute of Navigation and CIGTF 22nd Guidance Test Symposium*, Albuquerque, NM, June 2003, pp. 456-467.